

A fully spectral kinematic match

(design note)

1 A fully spectral kinematic match

We derive a model of droplet–bath contact in which the bath and droplet are represented as coupled spectral oscillator systems, following Alventosa, Cimpeanu & Harris (2023); contact is enforced over the entire pressed region and the pressure is solved for directly, following Agüero et al. (2026); and the map between the bath’s cylindrical coordinate and the droplet’s polar angle is used only in its explicit, forward direction. The section is organized as follows: §1.1 fixes notation and the geometric relation between the bath’s and droplet’s coordinates; §1.2 and §1.3 state the (unmodified) bath and droplet models; §1.4 derives, once, the affine dependence of each interface’s state on the pressure induced by implicit time-stepping; §1.5 derives the contact closure; §1.6 solves it as a one-dimensional outer iteration over the contact angle wrapping an inner solve for the pressure; §1.7 collects the complete, self-contained system solved at each time step; and §3 records arguments from earlier revisions of this document that were wrong, together with the structure removed along with them.

1.1 Governing equations and interface parametrization

We use the nondimensionalization of Alventosa, Cimpeanu & Harris (2023): length scale R (the undeformed droplet radius), time scale $t_\sigma = \sqrt{\rho R^3/\sigma}$, force scale $2\pi\sigma R$, giving the Weber, Bond and Ohnesorge numbers We, Bo, Oh as defined there. Cylindrical coordinates (r, z) describe the bath, with $z = 0$ the undisturbed free surface and gravity acting in $-\hat{z}$; spherical coordinates (ξ, θ) , sharing the same axis, are centred on the droplet’s centre of mass, with $\theta = 0$ along $-\hat{z}$ so that $\theta = 0$ is the point of the droplet nearest the bath. The bath interface height and the droplet radius are decomposed as

$$\eta(r, \tau) = \sum_{m=0}^M a_m(\tau) J_0(k_m r), \quad \xi(\theta, \tau) = 1 + \sum_{l=2}^L \beta_l(\tau) P_l(\cos \theta), \quad (1)$$

and the droplet’s centre of mass is at height $z_{cm}(\tau)$ above the undisturbed bath surface. A point on the droplet surface at polar angle θ lies, in the bath’s cylindrical coordinates, at horizontal distance and height

$$r(\theta, \tau) = \xi(\theta, \tau) \sin \theta, \quad z_d(\theta, \tau) = \xi(\theta, \tau) \cos \theta \quad (2)$$

below the centre of mass (so a point’s absolute height, in the bath’s z -coordinate, is $z_{cm}(\tau) - z_d(\theta, \tau)$: $z_d(0, \tau) = \xi(0, \tau) > 0$ places $\theta = 0$ below z_{cm} , consistent with it being the point nearest the bath), so that the droplet’s underside, at the same instant, meets the bath surface if and only if $\eta(r(\theta, \tau), \tau) = z_{cm}(\tau) - z_d(\theta, \tau)$. Equation (2) is a map from θ to (r, z_d) , given explicitly, for any θ , once $\{\beta_l(\tau)\}$ is known; it is never inverted for θ given r in what follows. We write $x := \cos \theta$ throughout when convenient, so that $P_l(\cos \theta) = P_l(x)$.

1.2 Bath interface model

The bath is governed by the linearized, weakly viscous free-surface equations of Alventosa, Cimpeanu & Harris (2023), themselves following Dias, Dyachenko & Zakharov (2008): with velocity potential $\phi(r, z, \tau)$,

$$\nabla_H^2 \phi + \partial_z^2 \phi = 0, \quad z \leq 0; \quad \partial_\tau \eta = \partial_z \phi + 2Oh \nabla^2 \eta, \quad z = 0; \quad (3)$$

$$\partial_\tau \phi = -Bo \eta - 2Oh \partial_z^2 \phi + \kappa[\eta] - p, \quad z = 0; \quad \partial_z \phi = 0 \text{ at } z = -h_0, \quad (4)$$

with $\nabla_H^2 = \partial_r^2 + r^{-1} \partial_r$ the horizontal Laplacian (written ∇^2 in the surface conditions above, where only the horizontal part acts) and $\kappa[\eta] = \nabla_H^2 \eta$ the linearized curvature and h_0 the (nondimensional) bath depth (at $z = -h_0$, consistent with $z \leq 0$; Alventosa, Cimpeanu & Harris 2023 print $z = h_0$), subject to the no-flux wall condition $\partial_r \eta = \partial_r \phi = 0$ at $r = b$. Expanding η as in (1) on the eigenfunctions of $(\nabla^2 + k_m^2)J_0(k_m r) = 0$ satisfying $J'_0(k_m b) = 0$, and eliminating ϕ between (3) and (4) mode by mode, gives one damped linear oscillator per bath mode,

$$\ddot{a}_m + 4Oh k_m^2 \dot{a}_m + (k_m^2 + Bo) k_m \tanh(k_m h_0) a_m = -2k_m \tanh(k_m h_0) c_m(\tau), \quad m = 0, \dots, M, \quad (5)$$

where

$$c_m(\tau) = \frac{2}{(bJ_0(k_m b))^2} \int_0^b p(r, \tau) r J_0(k_m r) dr \quad (6)$$

is the m -th Fourier–Bessel coefficient of the actual contact pressure p , extended by zero outside the pressed region. The normalization deserves a remark, because it differs from the one printed by Alventosa, Cimpeanu & Harris (2023) and the difference is not cosmetic. The no-flux wall condition gives $J'_0(k_m b) = 0$, equivalently $J_1(k_m b) = 0$, so $k_m b$ is the m -th zero of J_1 and the squared norm of the basis function is

$$\int_0^b J_0(k_m r)^2 r dr = \frac{b^2}{2} [J_0(k_m b)^2 + J_1(k_m b)^2] = \frac{b^2}{2} J_0(k_m b)^2, \quad (7)$$

whence the weight $2/(bJ_0(k_m b))^2$ used above. Alventosa, Cimpeanu & Harris (2023)’s eq. (300) instead prints $2/(bJ_1(k_m))^2$, which is the normalizer for a *Dirichlet* basis ($J_0(k_m b) = 0$, $k_m b$ a zero of J_0) and moreover evaluates J_1 at k_m rather than $k_m b$. Under the no-flux condition they themselves impose ($\partial_n \phi = 0$ on the walls, hence $\partial_r \eta = 0$ at $r = b$), $J_1(k_m b)$ vanishes identically, so their expression is inconsistent with their own basis. The discrepancy is mode-dependent and locally severe rather than a uniform rescaling: evaluated at $b = 6$ against (7), the ratio of their weight to the correct one ranges over roughly 0.16 to 1.8 for $m = 1, \dots, 8$ but reaches 116 at $m = 7$, because $k_7 = 3.7933$ falls close to the first zero of J_1 at 3.8317, driving $J_1(k_m) \rightarrow 0$ and their weight to diverge. An earlier revision of this document printed that ratio list inverted, which contradicted this very explanation — a diverging $J_1(k_m)^{-2}$ cannot produce a near-zero ratio in the stated orientation. An earlier revision of this document reproduced their expression verbatim; every bath-side quantity reported here uses (7).

1.2.1 Wall condition: a free versus a pinned contact line

Everything above assumes the bath meets the container wall at a free, 90° contact line: $\partial_r \eta = 0$ at $r = b$, the same condition the flow obeys. A second experimental configuration is of interest — a finite container with no-flux walls in which the free surface is *pinned* at the triple point, $\eta(b, \tau) = 0$ with the wall slope free, as a real meniscus anchored on a machined rim behaves. The two conditions

are not compatible with a single Fourier–Bessel set, and the incompatibility is structural rather than a matter of resolution.

The reason is not a limitation of any basis. It is a property of the continuum problem, and it is worth deriving because it determines which pinned models are admissible at all. Exact no-flux on a vertical wall means $\partial_r \phi(b, z) = 0$ for every $z \in [-h_0, 0]$; differentiating in z gives $\partial_r \partial_z \phi(b, 0) = 0$, and the kinematic condition (3) then yields

$$\partial_\tau [\partial_r \eta(b, \tau)] = \partial_r \partial_z \phi(b, 0) + 2Oh \partial_r \nabla^2 \eta(b, \tau) = O(Oh). \quad (8)$$

In a rigid right-cylindrical container with no-flux walls the free surface’s slope at the wall is therefore *frozen*, to $O(Oh)$, at whatever value it starts with. The consequence is sharper than an inconvenience: “pinned at the rim with a freely time-varying wall slope” is not a solution of (3)–(4) under no-flux walls at all — it is over-determined. The admissible pinned configuration is the *clamped edge*, $\eta(b) = 0$ together with $\partial_r \eta(b)$ frozen, and a constant nonzero wall slope is accommodated by splitting $\eta = \eta_s(r) + \sum_m a_m(\tau) J_0(k_m r)$ with η_s a static meniscus satisfying $\eta_s(b) = 0$, $\eta'_s(b) \neq 0$ and the dynamic part on the no-flux set. Pinning, no-flux and a nonzero wall slope are then all exact simultaneously. (The pinned-edge eigenvalue problem in a rigid container is classical: see Scott 1979 and Graham-Eagle 1983. Hocking 1987’s Robin edge condition, of the form $k J_1(kb) = \lambda J_0(kb)$, interpolates between the free and pinned limits as λ runs from 0 to ∞ and is a further option not pursued here.)

Against that background the two discrete routes are as follows. The no-flux set $\{J_0(k_m r) : J_1(k_m b) = 0\}$ used above has $\partial_r \eta(b) = 0$ for every member; the Dirichlet set $\{J_0(k_m r) : J_0(k_m b) = 0\}$ has $\eta(b) = 0$ for every member with the wall slope free, which is pinning as an identity of the basis, but its radial velocity at the wall goes like $J_1(k_m b)$, which does not vanish.

Route A: the Dirichlet basis. Take $k_m b$ to be the m -th zero of J_0 . Pinning is then exact mode by mode — an identity of the basis, not a constraint to be enforced — and the wall slope is unconstrained. Two consequences follow. First, the squared norm (7) becomes

$$\int_0^b J_0(k_m r)^2 r dr = \frac{b^2}{2} [J_0(k_m b)^2 + J_1(k_m b)^2] = \frac{b^2}{2} J_1(k_m b)^2, \quad (9)$$

so the Fourier–Bessel weight in (6) is $2/(b J_1(k_m b))^2$. This is the Dirichlet normalizer whose form Alventosa, Cimpeanu & Harris (2023) print — modulo the second, independent error §1.2 identifies, since they evaluate J_1 at k_m rather than at $k_m b$, and $b = 6$ here. It is the right normalizer for a pinned bath applied to a free one, which is part of why the slip was easy to make; the two configurations also differ in the eigenvalues themselves and in the presence of the $k = 0$ mode, so the substitution is not the whole of the difference.

Second, the piston mode disappears. The zeros of J_0 begin at $k_1 b = 2.4048$, so there is no $k = 0$ member, whereas the no-flux set begins with $k_0 = 0$ and needs it. This is not a convention: a surface pinned on its entire boundary cannot translate uniformly, so the uniform mode is not an eigenfunction of the pinned problem and its absence is the correct physics.

The cost is severe, and stating it as “a leaking wall” understates it. What the Dirichlet basis loses is *exact conservation of bath volume*, which the no-flux basis possesses structurally. For the no-flux set every non-piston mode is volume-neutral, since

$$\int_0^b J_0(k_m r) r dr = \frac{b}{k_m} J_1(k_m b) = 0 \quad \text{whenever } J_1(k_m b) = 0, \quad (10)$$

and the only volume-carrying mode is the piston, whose pressure response κ_0 vanishes identically, so it can never be driven. Displaced volume is thus conserved by construction rather than by accuracy.

For the Dirichlet set (10) fails for every mode and nothing constrains the sum. Measured over the reference impact, $|\int_0^b \eta r dr|$ stays at 5×10^{-15} under `:free` but reaches 2.79 under `:pinned` — 17.5 droplet radii cubed of bath volume created from nothing, four times the droplet’s own volume of $4\pi/3$. For a model whose entire purpose is to transfer displaced volume from droplet to bath and radiate it as capillary waves, that is disqualifying, not a caveat.

Route B: the no-flux basis with a constraint. Retain the Neumann set, so that no-flux and (10) both hold identically, and impose pinning as the single scalar constraint $\mathcal{C}(a) = \sum_m a_m J_0(k_m b) = 0$ carried by a multiplier Λ — physically the line force the rim exerts on the contact line, which is a quantity of interest in its own right rather than an artifice. The multiplier enters as a line force $p_\Lambda = \Lambda \delta(r - b)$, whose Fourier–Bessel coefficient under (6) is $c_m^\Lambda = 2\Lambda/(bJ_0(k_m b))$, taking $\int_0^b f(r)\delta(r-b) dr = f(b)$ at the domain’s own boundary — a convention, since a delta sitting exactly at an endpoint admits no unique weight, but a harmless one, as any other choice only rescales Λ itself rather than changing the structure below. So (22) gains one term and the constraint closes on

$$a_m = \alpha_m + \kappa_m c_m + \frac{2}{b} \frac{\kappa_m \Lambda}{J_0(k_m b)}, \quad D = \frac{2}{b} \sum_m \kappa_m. \quad (11)$$

The $J_0(k_m b)$ factors cancel out of D entirely. Since $\kappa_m < 0$ for every non-piston mode (the sole exception, $\kappa_0 = 0$, contributes nothing to the sum either way) the sum cannot cancel, and it converges because $\kappa_m \sim -2/k_m^2$; the elimination leaves a_m affine in c with a rank-one correction to the bath compliance, so the inner Galerkin system stays square and linear in the same unknowns and the nested closure of §1.5 is untouched, at a cost of one $O(M)$ dot product. Two further properties matter. The rank-one correction is self-adjoint in the same weighted pairing $\langle \cdot, \cdot \rangle_w$, $w_m \propto J_0(k_m b)^2$, that §1.5.5 uses — measured at relative asymmetry 1.1×10^{-17} , matching that section’s own 10^{-16} figure for the free operator, not merely asserted — so it does not disturb the self-adjointness the formulation was built around. And $\kappa_0 = 0$, so the rim force cannot drive the piston mode: pinning by constraint conserves volume exactly, and does not reintroduce Route A’s defect.

Three further facts, assembled and checked at operator level rather than argued in isolation (`derivations/feasibility_pinned_contact_line.jl`, §5), close the remaining gaps between the algebra above and an implementation. First, $J_0(k_m b)$ never vanishes on the Neumann set: $J'_0 = -J_1$, so a common zero of J_0 and J_1 would force J_0 to vanish to second order there, which for a solution of Bessel’s equation forces $J_0 \equiv 0$ — false. $|J_0(k_m b)|$ does decay with m , but along the generic $m^{-1/2}$ Bessel envelope rather than toward a near-miss with a zero, and the quantity that actually appears in a_m , $\kappa_m/J_0(k_m b)$, falls as $m^{-3/2}$ and saturates (its maximum over m stops growing by $M \sim 640$): no per-mode term is amplified by refining the truncation. Second, the constraint is an identity of the assembled $(M+1) \times (M+1)$ operator $\mathcal{A}_{\text{clamped}}$, not a property of one solved-for c : $\sum_m (\mathcal{A}_{\text{clamped}})_{mn} J_0(k_m b) = 0$ for every column n , measured at 5×10^{-22} . Third, the sign is preserved, and not merely at the one parameter point checked above: writing $-W\mathcal{A}_{\text{clamped}} = \text{diag}(-\kappa_m J_0(k_m b)^2) + (\frac{-2}{bD})zz^\top$ with $z_m = \kappa_m J_0(k_m b)$, both terms are positive semidefinite given only $\kappa_m \leq 0$ for every mode and $D < 0$ — and those two facts hold for *any* admissible step, since $\alpha = (1 + 2s)/(1 + s) > 0$ for every BDF2 step ratio $s > 0$ and $Oh, Bo \geq 0$ in κ_m ’s closed form. The rank-one term therefore cannot push $-\mathcal{A}_{\text{clamped}}$ out of positive semidefiniteness at any step, not only the representative one measured numerically above.

Route B’s representational limitation is far weaker than an earlier revision of this section claimed. Expanding the pinned target $\eta = J_0(j_{0,1}r/b)$ on the Neumann set, the truncation error computed analytically by Parseval falls as $M^{-3/2}$, and the wall slope is reproduced pointwise at every $r < b$; only at the single point $r = b$ does the truncated sum have zero derivative, with

a boundary layer of width $\sim b/k_M$ and a bounded Gibbs overshoot of a few percent. The failure is confined to a set of measure zero. Given (8) there is in any case no time-varying wall slope to represent.

That earlier revision reached the opposite conclusion from a diagnostic that was measuring its own grid. It finite-differenced the fitted profile across the last interval of a fixed 4000-point grid, i.e. *inside* the boundary layer, so the number returned scales as $\Delta r k_M$ rather than as any representation error: holding $M = 640$ fixed and refining the grid drives it from -0.033 to -0.0005 , toward zero, while the true slope is -0.208 . The suspiciously clean doubling of that “slope” with each doubling of M was the signature of the artifact. The relative L^2 column was likewise quadrature-limited and could not discriminate. Both are corrected in `derivations/feasibility_pinned_contact_line.jl`, which now differentiates the series analytically at fixed $r < b$.

Route A is what the implementation provides as a non-default diagnostic option (`wall=:pinned`), not a physical model: pinning is exact and sustained — $|\eta(b, \tau)|$ stays at roundoff throughout an impact, and both weights agree with $1/\|J_0(k_m \cdot)\|^2$ to better than 10^{-15} — but volume is not conserved, by the margin quoted above. Route B, the recommended formulation consistent with (3)–(4), is implemented as `wall=:clamped`, applying (11) at every step, including free flight (pinning is a constraint on a_m itself, not a byproduct of contact pressure). Measured over the same reference impact, `:clamped` holds both quantities at roundoff simultaneously: $|\eta(b, \tau)|$ at 7×10^{-17} and $|\int_0^b \eta r dr|$ at 2×10^{-14} , where `:pinned` traded one for the other. Two scope limitations should be recorded. `:clamped` reuses the `:free` basis and weight exactly, and §5 of `derivations/feasibility_pinned_contact_line.jl` verifies the corrected response operator’s self-adjointness and sign at the operator level (§1.2.1 above), but the step-level diagnostics of §1.5 and §1.5.5 — conditioning of the assembled nonlinear system, the resolvable pressure dimension, the tangency root — are free-wall measurements that have not been repeated for `:clamped` at a real contact step. The difference between `:free` and `:clamped` is now measurable rather than an artifact, since both conserve volume, and it is strongly b -dependent: at the production default $b = 6$ it is negligible ($\Delta\text{CoR} = -0.001$), but shrinking the container moves it far outside step-controller noise and it is *not* monotone in b — $\Delta\text{CoR} = +0.26$ at $b = 4$, $+0.11$ at $b = 3$, $+0.07$ at $b = 2$, -0.21 at $b = 1.5$ — consistent with a container resonance crossing the impact timescale as the eigenvalues $k_m \propto j_{1,m}/b$ move with b . The wall condition is therefore a real effect for a small or moderately sized bath, and an irrelevant one at $b = 6$; neither the resolvable-dimension diagnostics above nor a convergence study in b itself have been repeated at these smaller radii, so the numbers are a gate, not a validated sweep.

1.3 Droplet interface model

The droplet’s small-amplitude oscillations about its spherical equilibrium are governed, for a general contact pressure $p(\theta, \tau)$, by the energy balance $\dot{T} = \dot{W} - \varepsilon$ of Alventosa, Cimpeanu & Harris (2023) (their eq. 3.7), with T the sum of kinetic and surface energy of the decomposition (1), $\dot{W}$ the rate of working of p on the surface, and ε the viscous dissipation of Lamb (1932). Substituting (1) and using the orthogonality of the $P_l(\cos \theta)$ on $[-1, 1]$ gives, mode by mode,

$$\ddot{\beta}_l + 2Oh(2l+1)(l-1)\dot{\beta}_l + l(l-1)(l+2)\beta_l = -(2l+1)l b_l(\tau), \quad l = 2, \dots, L, \quad (12)$$

where

$$b_l(\tau) = \int_{-1}^1 p(x, \tau) P_l(x) dx \quad (13)$$

is the un-normalized Legendre projection of the actual pressure, in the convention Alventosa, Cimpeanu & Harris (2023) use for the projection c_l of their prescribed shape; here b_l plays the role

their product $f(\tau)c_l(\tau)$ played.

The two parent models print forcings for (12) that differ by $(2l+1)^2$, so a remark on which is adopted, and why, is warranted. Agüero et al. (2026) write the forcing as $-l\mathcal{B}_l$ with $\mathcal{B}_l$ the coefficient in $p = \sum_l \mathcal{B}_l P_l(\cos\theta)$, but state its inverse transform as $\mathcal{B}_l = (2l+1)^{-1} \int p P_l dx$; since $\int_{-1}^1 P_l^2 dx = 2/(2l+1)$, the inverse of their own expansion is $\mathcal{B}_l = \frac{1}{2}(2l+1) \int p P_l dx$, larger by $(2l+1)^2/2$. Their mode equation is correct under the latter reading, and an independent derivation settles which: for $\xi = R + \sum_l A_l P_l$ the kinetic and surface energies are $2\pi\rho R^3 \dot{A}_l^2/(l(2l+1))$ and $2\pi\sigma(l-1)(l+2)A_l^2/(2l+1)$, and the rate of working of p is $-2\pi R^2 \dot{A}_l b_l$, so $\frac{d}{d\tau}[T + E_s] = \dot{W}$ gives $\rho R^3 \ddot{A}_l = -\sigma l(l-1)(l+2)A_l - \frac{1}{2}R^2 l(2l+1)b_l$. This reproduces Alventosa, Cimpeanu & Harris (2023)'s dimensional equation coefficient for coefficient, and matches Agüero et al. (2026)'s only with $\mathcal{B}_l = \frac{1}{2}(2l+1)b_l$, confirming that the discrepancy is a misprinted projection formula in the latter rather than a difference of physics — in particular it is not a lab-frame versus drop-frame difference, since both papers refer the deformation modes to the droplet's own centre of mass, and a change of frame is a rigid translation that projects entirely onto $l=1$. The residual factor of two between (12) and the derivation above is the pressure scale: (12) and (14) are jointly consistent with $2\sigma/R$, matching Alventosa, Cimpeanu & Harris (2023)'s force scale $2\pi\sigma R$ and their normalization $f = 2 \int_0^{r_c} p r dr$. Modes $l=0$ (excluded by incompressibility of the droplet) and $l=1$ do not appear in (1); $l=1$ is instead governed by the motion of the centre of mass,

$$\dot{z}_{cm} = v, \quad \dot{v} = \frac{3}{2}f(\tau) - Bo, \quad f(\tau) = 2 \int_0^{r_c(\tau)} p(r, \tau) r dr, \quad (14)$$

with $r_c(\tau)$ the (as yet undetermined) contact radius. Under the ansatz $p = f H_r(r/r_c)$ of Alventosa, Cimpeanu & Harris (2023), whose normalization $\int_0^{r_c} H_r(r/r_c) r dr = \frac{1}{2}$ they impose, $2 \int_0^{r_c} p r dr = 2f \int_0^{r_c} H_r(r/r_c) r dr = 2f \cdot \frac{1}{2} = f$, so (14) reduces to their own definition of the net contact force f when the shape is prescribed, and generalizes it when it is not.

1.4 Time discretization: affine response to pressure

Equations (5), (12) and the pair (14) are each, mode by mode, of the form

$$\ddot{x} + 2\gamma\dot{x} + \omega^2 x = F u(\tau), \quad (15)$$

with x one of a_m, β_l, z_{cm} , u the corresponding pressure moment c_m, b_l, f , and (γ, ω^2, F) read off from (5), (12), (14) respectively:

$$(\gamma, \omega^2, F)_{\text{bath}} = \left(2Oh k_m^2, (k_m^2 + Bo)k_m \tanh(k_m h_0), -2k_m \tanh(k_m h_0) \right), \quad (16)$$

$$(\gamma, \omega^2, F)_{\text{drop}} = \left(Oh(2l+1)(l-1), l(l-1)(l+2), -(2l+1)l \right), \quad (\gamma, \omega^2, F)_{cm} = \left(0, 0, \frac{3}{2} \right). \quad (17)$$

We discretize (15) in time with the variable-step BDF2 scheme of Agüero et al. (2026): with $s^k = \delta_t^k / \delta_t^{k-1}$, $\mathbf{a}^k = (1 + 2s^k)/(1 + s^k)$, $b^k = -(1 + s^k)$, $c^k = (s^k)^2/(1 + s^k)$, and $\dot{f}^{k+1} \approx (\mathbf{a}^k f^{k+1} + b^k f^k + c^k f^{k-1})/\delta_t^k$ for any variable f , the first-order system $(x, y=\dot{x})$ becomes

$$\mathbf{a}x^{k+1} + bx^k + cx^{k-1} = \delta y^{k+1}, \quad \mathbf{a}y^{k+1} + by^k + cy^{k-1} = \delta \left(-2\gamma y^{k+1} - \omega^2 x^{k+1} + F u^{k+1} \right), \quad (18)$$

where $\delta := \delta_t^k$ and the step superscript on $\mathbf{a}, b, c$ is dropped for brevity. Solving the first equation of (18) for $y^{k+1} = (\mathbf{a}x^{k+1} + bx^k + cx^{k-1})/\delta$ and substituting into the second gives

$$\frac{\mathbf{a}}{\delta} \left(\mathbf{a}x^{k+1} + bx^k + cx^{k-1} \right) + by^k + cy^{k-1} = -2\gamma \left(\mathbf{a}x^{k+1} + bx^k + cx^{k-1} \right) - \delta \omega^2 x^{k+1} + \delta F u^{k+1}, \quad (19)$$

and collecting every term proportional to x^{k+1} on the left,

$$x^{k+1} \left[\frac{\mathfrak{a}^2}{\delta} + 2\gamma\mathfrak{a} + \delta\omega^2 \right] = \delta F u^{k+1} - \frac{\mathfrak{a}}{\delta} (bx^k + cx^{k-1}) - 2\gamma(bx^k + cx^{k-1}) - by^k - cy^{k-1}, \quad (20)$$

so that, multiplying through by δ ,

$$x^{k+1} = (\text{known history}) + \kappa u^{k+1}, \quad \kappa = \frac{\delta^2 F}{\mathfrak{a}^2 + 2\gamma\mathfrak{a}\delta + \delta^2\omega^2}. \quad (21)$$

Substituting (16)–(17) into (21) gives, explicitly,

$$a_m^{k+1} = \alpha_m^{k+1} + \kappa_m c_m^{k+1}, \quad \kappa_m = \frac{-2\delta^2 k_m \tanh(k_m h_0)}{\mathfrak{a}(\mathfrak{a} + 4\delta Oh k_m^2) + \delta^2(k_m^2 + Bo)k_m \tanh(k_m h_0)}, \quad (22)$$

$$\beta_l^{k+1} = \gamma_l^{k+1} + \lambda_l b_l^{k+1}, \quad \lambda_l = \frac{-\delta^2(2l+1)l}{\mathfrak{a}(\mathfrak{a} + 2\delta Oh(2l+1)(l-1)) + \delta^2 l(l-1)(l+2)}, \quad (23)$$

$$z_{cm}^{k+1} = \mu^{k+1} + \kappa_{cm} f^{k+1}, \quad \kappa_{cm} = \frac{3\delta^2}{2\mathfrak{a}^2}, \quad (24)$$

with $\alpha_m^{k+1}, \gamma_l^{k+1}, \mu^{k+1}$ depending only on $\delta, \mathfrak{a}, b, c$ and the two preceding time levels. Each of $a_m^{k+1}, \beta_l^{k+1}, z_{cm}^{k+1}$ is therefore an affine function of the pressure moment forcing it at the same time level; the slopes $\kappa_m, \lambda_l, \kappa_{cm}$ are known once $\delta, \mathfrak{a}$ are fixed by the time step, and it is this affine dependence that closes the contact problem below into a purely algebraic one at each time step.

1.5 Modeling contact

We now determine the pressure and the contact angle $\theta_c(\tau)$ so that the bath and droplet surfaces coincide throughout $[0, \theta_c(\tau)]$.

1.5.1 Pressure representation

We represent the pressure as a polynomial of degree N in $x = \cos\theta$, compactly supported on the pressed region by construction. Two representations are available and both are viable. A nonzero polynomial (or any convergent real-analytic series) cannot vanish identically on a sub-interval without vanishing identically everywhere — a degree- N polynomial has at most N roots, so forcing it to zero across a continuum forces every coefficient to zero — so a representation over the full domain $x \in [-1, 1]$ with “zero outside contact” imposed as a separate weak condition can only approximate that property, never satisfy it exactly. An earlier revision of this document treated that observation as disqualifying, and concluded that piecewise support was a *structural requirement* of a genuinely spectral pressure unknown. That conclusion was wrong, and is withdrawn: weak enforcement of $p \equiv 0$ outside contact has exactly the standing every other Galerkin condition in this document has, including (27) itself, which likewise cannot enforce its own pointwise condition exactly. Rejecting the global representation on grounds that condemn the rest of the formulation equally was a double standard, not an argument.

We nonetheless adopt the piecewise representation, for a reason that survives scrutiny: with the contact angle determined by the outer scalar iteration of §1.5.6 rather than carried as an unknown alongside the pressure coefficients, the support endpoint x_c is a fixed *parameter* throughout each inner solve. Exact vanishing outside contact is then free — it costs no extra equations, introduces

no approximation, and leaves nothing to be checked afterward — so there is no reason to trade it for the weaker global alternative.

Within that constraint, the basis functions themselves are not required to be monomials, and monomials are in fact a poor choice: writing $\psi(x, \tau) := (x - x_c(\tau))/(1 - x_c(\tau)) \in [0, 1]$ for the pressure's own rescaled coordinate, the monomials ψ^n have a self-overlap (Gram) matrix $\int_0^1 \psi^n \psi^m d\psi = 1/(n + m + 1)$ — the classical Hilbert matrix, whose condition number grows like 1.6×10^4 already at $N = 3$, independent of x_c . We instead use the shifted Legendre polynomials $\tilde{P}_n(\psi) := P_n(2\psi - 1)$, exactly orthogonal under the same plain- L^2 inner product used throughout this document (the same one Gauss–Legendre quadrature and the droplet's own $P_l(x)$ basis already use, unlike Chebyshev polynomials, whose optimality is with respect to a different, weighted inner product that does not transfer to the mixed projections computed below), giving

$$p(x, \tau) = \sum_{n=0}^N \hat{c}_n(\tau) \tilde{P}_n(\psi(x, \tau)), \quad x \geq x_c(\tau); \quad p(x, \tau) \equiv 0, \quad x < x_c(\tau), \quad x_c(\tau) := \cos \theta_c(\tau). \quad (25)$$

$\tilde{P}_n(\psi(x, \tau))$ is, for every n , exactly a degree- n polynomial in x (an affine change of variable composed with a degree- n polynomial), so (25) has exactly the same total polynomial degree N as the monomial basis it replaces — confirmed symbolically, not merely asserted, in `derivations/verify_legendre_pressure_`. Their self-Gram matrix is, by orthogonality, diagonal: $\text{cond} = 2N + 1$ at every N and every x_c , against 1.6×10^4 for the monomial basis at $N = 3$.

1.5.2 Full contact

Contact over the entire pressed region requires

$$\eta(r(\theta, \tau), \tau) = z_{cm}(\tau) - z_d(\theta, \tau), \quad 0 \leq \theta \leq \theta_c(\tau), \quad (26)$$

with η , z_{cm} , z_d affine in c_m , f , b_l respectively by (22)–(24). The $N + 1$ coefficients $\hat{c}_n(\tau)$ cannot satisfy (26) at every $\theta \in [0, \theta_c(\tau)]$; we impose it in the weighted-residual sense obtained by projecting the mismatch onto the $N + 1$ polynomials used in (25),

$$\int_{x_c}^1 \left[\eta(r(\theta, \tau), \tau) - z_{cm}(\tau) + z_d(\theta, \tau) \right] \tilde{P}_n(\psi(x, \tau)) w(x, \tau) dx = 0, \quad n = 0, \dots, N, \quad (27)$$

$N + 1$ equations. By (23), $\beta_l(\tau) = \gamma_l(\tau) + \lambda_l b_l(\tau)$, and $b_l(\tau) = \int_{x_c}^1 p(x, \tau) P_l(x) dx$ depends on $\hat{c}_n(\tau)$ alone — not on η , r , or z_d — so $\beta_l(\tau)$ is an explicit function of the same unknowns (27) is solving for. Substituting it makes (27) genuinely nonlinear in $\hat{c}_n(\tau)$: $\beta_l(\tau)$ enters ξ , hence $r(\theta, \tau)$, hence $J_0(k_m r(\theta, \tau))$ inside η (through $c_m(\tau)$, §1.5.4 below), and J_0 composed with an affine function of $\hat{c}_n(\tau)$ is not itself affine. (27) is solved for $\hat{c}_n(\tau)$ at a fixed trial contact angle by the inner nonlinear solve of §1.6, with $\theta_c(\tau)$ itself fixed by the outer scalar selection of §1.5.6; we first evaluate the four projections entering (27) and (6) through z_d , z_{cm} and η , each as an explicit function of $\hat{c}_n(\tau)$ and $\theta_c(\tau)$, in turn.

1.5.3 The polynomial projections

The centre-of-mass and droplet-surface contributions to (14), (13) and (27) involve only ξ and p , hence only polynomials in x , and are therefore *exactly* (not merely approximately) evaluable by Gauss–Legendre quadrature (30), given enough nodes: an n_q -point rule is exact for any polynomial of degree at most $2n_q - 1$, and every integrand here has an explicitly known, finite degree.

This replaces a closed-form antiderivative recursion (viable only for the monomial basis (25) previously used here, via the Legendre antiderivatives of Agüero et al. (2026)’s App. A) with the same quadrature machinery already required below for the bath’s J_0 terms, which have no antiderivative in closed form regardless of basis — one uniform mechanism for every projection in this document, not two.

$b_l(\tau) = \int_{x_c}^1 p(x, \tau) P_l(x) dx$ of (13) has integrand $\tilde{P}_n(\psi(x, \tau)) P_l(x)$, degree $N + L$ in x (degree N from (25), degree L from P_l), hence exact for $n_q \geq \lceil (N + L + 1)/2 \rceil$ quadrature points. The droplet contribution to (27),

$$\int_{x_c}^1 z_d(\theta, \tau) \tilde{P}_n(\psi(x, \tau)) dx = \int_{x_c}^1 x \left[1 + \sum_{l=2}^L \beta_l(\tau) P_l(x) \right] \tilde{P}_n(\psi(x, \tau)) dx, \quad (28)$$

has integrand degree $N + L + 1$ (the extra factor of x in $z_d = \xi \cos \theta$), exact for $n_q \geq \lceil (N + L + 2)/2 \rceil$. Both degree counts are confirmed symbolically, not hand-counted, in `derivations/verify_legendre_pressure_ba` together with the degree- n claim for $\tilde{P}_n(\psi(x, \tau))$ itself.

The centre-of-mass integral in (14) is likewise a pure polynomial integral: since $r(x, \tau)^2 = \xi(x, \tau)^2(1 - x^2)$ is itself a polynomial in x (the $\sqrt{1 - x^2}$ in $\sin \theta$ cancels once squared), $2 \int_0^{r_c} p r dr = - \int_{x_c}^1 p(x, \tau) d[r(x, \tau)^2]$ has integrand degree $N + 2L + 1$, exact for $n_q \geq \lceil (N + 2L + 2)/2 \rceil$ — unchanged from the monomial basis, since $p(x, \tau)$ still has total degree N regardless of which degree- N polynomial family represents it. The minus sign is not optional: $\theta = 0$ (the pole, $r = 0$) corresponds to $x = 1$, and $\theta = \theta_c$ (the contact edge, $r = r_c$) corresponds to $x = x_c < 1$, so r *increases* as x *decreases* over the contact patch; substituting $r \rightarrow x$ in $\int_0^{r_c} (\cdot) d[r^2]$ and reorienting the limits from $(1, x_c)$ to the natural $(x_c, 1)$ introduces exactly this sign. Omitting it makes the pressure force accelerate the droplet further into the bath rather than decelerating it — confirmed by evaluating both sides directly for a constant test pressure $p \equiv 1$, where the unsigned form returns $-r_c^2$ against the correct $2 \int_0^{r_c} r dr = +r_c^2$.

1.5.4 The quadrature projections

The bath-side projection c_m of (6) and the bath contribution to (27) both involve $J_0(k_m r(x, \tau))$, and J_0 composed with the non-polynomial map $r(x, \tau) = \xi(\arccos x, \tau) \sqrt{1 - x^2}$ has no elementary antiderivative. We evaluate

$$c_m(\tau) = \frac{2}{(b J_0(k_m b))^2} \int_{x_c}^1 p(x, \tau) J_0(k_m r(x, \tau)) r(x, \tau) \left| \frac{dr}{dx} \right| dx, \quad W_n^{(m)}(\tau) := \int_{x_c}^1 J_0(k_m r(x, \tau)) \tilde{P}_n(\psi(x, \tau)) dx \quad (29)$$

by n_q -point Gauss–Legendre quadrature on $[x_c, 1]$: with $\{s_i, w_i\}_{i=1}^{n_q}$ the standard nodes and weights on $[-1, 1]$ and the affine map $x = x_c + (1 + s)(1 - x_c)/2$,

$$\int_{x_c}^1 g(x) dx \approx \frac{1 - x_c}{2} \sum_{i=1}^{n_q} w_i g\left(x_c + \frac{(1 + s_i)(1 - x_c)}{2}\right), \quad (30)$$

applied to g equal to the integrand of c_m and, separately, of $W_n^{(m)}$. No root is found in evaluating (30): $r(x, \tau)$ is a forward evaluation of (2) at each quadrature abscissa, not an inversion, and no unknown of the problem is attached to an abscissa; (29) is one definite integral per (m, n) , feeding linearly into (5) and (27) respectively, in exactly the sense that any spectral method evaluates a forcing term that is not itself expressible in the trial basis by quadrature. With (29), the bath contribution to (27) is $\sum_m a_m(\tau) W_n^{(m)}(\tau)$.

1.5.5 The compliance operator, and what the contact conditions determine

Before closing the system it is worth asking what object the closure is inverting, because its structure decides both what is determinable and how well conditioned the determination is. Collect the pressure-dependent part of the contact mismatch into a single linear map. By (22)–(24) each of η, z_{cm}, z_d responds affinely to the pressure moments, and by (6), (13) and (14) each moment is a linear functional of p ; composing, at frozen geometry,

$$C(x, \tau) = C_{\text{free}}(x, \tau) + (\mathcal{A}p)(x, \tau), \quad (\mathcal{A}p)(x) = \int_{x_c}^1 \mathcal{K}(x, x') p(x') dx', \quad (31)$$

with C_{free} the mismatch the BDF2 history alone would produce and $\mathcal{K}$ assembled from three contributions,

$$\mathcal{K} = \underbrace{\sum_m \kappa_m \frac{2}{(b J_0(k_m b))^2} J_0(k_m r(x)) J_0(k_m r(x')) w(x')}_{\text{bath}} \underbrace{- 2\kappa_{cm} w(x')}_{\text{centre of mass}} + \underbrace{\sum_l \lambda_l x P_l(x) P_l(x')}_{\text{droplet}}, \quad (32)$$

where $w(x) := -\frac{1}{2} d[r^2]/dx$, which equals $r|dr/dx|$ and hence supplies the cylindrical area element $r dr = w dx$ provided $dr/dx \leq 0$ across the patch — a genuine restriction on the admissible contact extent, not an identity, recorded as such in §1.5.7. $\mathcal{A}$ is the *compliance operator* of the contact problem: it maps a trial pressure to the gap displacement that pressure induces. Its properties, measured in `derivations/audit_compliance_operator.jl` and `derivations/audit_weak_determinacy.jl`, govern what follows, and several of them are less favourable than an earlier revision of this section claimed.

$\mathcal{A}$ is self-adjoint under a consistent choice of pairing, and that choice is adopted. Self-adjointness of $\mathcal{A}$ with respect to $\langle u, v \rangle_\mu = \int uv \mu dx$ requires $\mu(x)\mathcal{K}(x, x') = \mu(x')\mathcal{K}(x', x)$. Read off (32): the bath and centre-of-mass terms carry their entire x' -dependence in the factor $w(x')$, so they are self-adjoint precisely for $\mu \propto w$ — confirmed at relative Frobenius asymmetry $\sim 10^{-16}$ at every θ_c tested. The droplet term as written, $\sum_l \lambda_l x P_l(x) P_l(x')$, is not: it would require $\mu \propto 1/x$ instead, and no weight satisfies both.

The obstruction is not, however, a property of the physics; it is a bookkeeping inconsistency, and it is the superposition of exactly two mismatches. First, c_m and f pair the pressure against the cylindrical element $w dx = r dr$ while b_l (13) pairs it against the spherical element $dx = \sin \theta d\theta$. Second, b_l pairs against the droplet’s *radial* displacement, following Alventosa, Cimpeanu & Harris (2023)’s energy balance, while the same mode returns to the contact mismatch through its *vertical* component $x\xi$. Repairing either alone leaves the asymmetry non-zero; repairing *both*,

$$b_l(\tau) = \int_{x_c}^1 p(x, \tau) x P_l(x) w(x, \tau) dx, \quad (33)$$

makes the droplet block — and hence the whole of $\mathcal{A}$ — self-adjoint in $\langle \cdot, \cdot \rangle_w$ to 10^{-16} at every deformation tested ($\beta_2 = 0, 0.02, 0.05, 0.10$), against $3.6\text{--}5.5 \times 10^{-3}$ for (13). Both mismatches must be repaired together: taking the vertical factor alone leaves $1.8\text{--}3.6 \times 10^{-3}$ and the measure alone leaves $1.7\text{--}1.8 \times 10^{-3}$. An earlier revision of this section tested them only in isolation, concluded that neither was “the” mechanism, and inferred that self-adjointness was unattainable; that inference was wrong and is withdrawn (§3).

Equation (33) is adopted in place of (13). It is a change of *linearisation convention*, not of physics, and it is $O(\theta_c^2)$: the pressure acts on the droplet surface along its true normal, which is neither exactly radial nor exactly vertical, and the two conventions differ only at second order in

the contact angle. Three points make the choice the better one. It is internally consistent, treating the droplet’s displacement in the same direction and against the same measure as the contact condition and the bath and centre-of-mass channels already use. It buys exact energy reciprocity of the coupled system, hence an energy functional and the monotone Galerkin convergence estimates that accompany one. And it degenerates to the parents’ convention exactly in the limit the parents work in: $x \rightarrow 1$ and $w \rightarrow 1$ as $\theta_c \rightarrow 0$, so (33) reduces to (13) at pole contact, which is precisely where Alventosa, Cimpeanu & Harris (2023) enforce their single-point match. Only Agüero et al. (2026) carry the inconsistency over a finite patch; Alventosa, Cimpeanu & Harris (2023) do not incur it at all, so no appeal to inheritance is needed to justify departing from it.

One consequence is used below. With (33) in force, and with the contact-region Galerkin conditions tested against the *same* basis that represents the pressure and in the *same* measure $w dx$, the inner Galerkin matrix is symmetric — measured at $\|G - G^\top\|/\|G + G^\top\| = 2.1, 1.9, 1.3 \times 10^{-16}$ at $\theta_c = 0.10, 0.20, 0.40$. Both conditions are necessary: pairing a different test basis against the trial basis destroys the symmetry even when the operator itself is self-adjoint, which is how the effect was first missed.

The symmetric part is positive semidefinite, not definite, and marginally so at larger patches. Measured eigenvalues of the symmetrised $-\mathcal{A}$ in the w -measure, $n_q = 40$: at $\theta_c = 0.05, 0.15, 0.30, 0.60$ the ratio $\lambda_{\min}/\lambda_{\max}$ is $-2.3 \times 10^{-16}, -1.2 \times 10^{-16}, -1.9 \times 10^{-12}, -4.2 \times 10^{-7}$, with 35, 31, 26, 18 of 40 eigenvalues below $10^{-12}\lambda_{\max}$. So the operator is semidefinite with a nullspace occupying roughly half the discretisation or more, and at $\theta_c = 0.60$ the most negative eigenvalue is too large relative to $\lambda_{\max}$ to attribute to the $h = 10^{-7}$ finite difference used for $d[r^2]/dx$ — semidefiniteness is not cleanly established there. What semidefiniteness does buy is uniqueness of the gap response $\mathcal{A}p$; it does *not* buy uniqueness of p , and an earlier revision claimed the latter. Nor is a Frobenius-norm comparison of symmetric and antisymmetric parts evidence of definiteness in either direction, as that revision also supposed. This is not in tension with the compactness discussed below: a large numerical nullspace is precisely the finite-dimensional face of an operator whose singular values accumulate at zero.

The $O(\delta^2)$ pressure sensitivity is a near-uniform scalar factor only in a restricted regime. Every term of (32) carries exactly one factor of κ_m, κ_{cm} or λ_l , each $O(\delta^2)$ by (21), which suggests $\mathcal{A} = \delta^2 \hat{\mathcal{A}}$ with $\hat{\mathcal{A}}$ independent of δ . But the denominators of (22)–(23) carry δ -dependent, *mode-dependent* terms, so the factorisation is only asymptotic, and the regime in which it holds is narrower than an earlier revision of this section asserted. The relative deviation of κ_m/δ^2 from its $\delta \rightarrow 0$ limit, maximised over m , is 0.011 at $M = 40, \delta = 10^{-3}$, but 0.273 at $M = 160$ and 0.960 at $M = 700$; for λ_l the corresponding figures are 0.052, 0.691 and 0.994. Uniformity therefore requires $\delta k_M^2 \ll 1$ and $\delta^2 L^3 \ll 1$, and is *violated* at the mode counts used in the resolution study below. Correspondingly $\|\mathcal{A}\|/\delta^2$ is measured at $5.97 \times 10^4, 1.51 \times 10^5, 1.60 \times 10^5, 1.60 \times 10^5$ across $\delta = 10^{-2}, \dots, 10^{-5}$ — settling, but varying by a factor of 2.5 at the largest step, which the earlier revision suppressed by quoting only the limit. The important point is that the refutation of the deleted acceleration-level closure (§3) never needed uniformity: it needs only $\text{cond}(c\mathcal{A}) = \text{cond}(\mathcal{A})$ together with the *measured* δ -insensitivity of the inner system, which is flat and of order ten (§1.5.6) and is a direct measurement rather than an inference.

$\mathcal{A}$ is compact, and its resolvable dimension must be measured rather than predicted. $\mathcal{A}$ is an integral operator with a smooth kernel, hence compact, hence has singular values accumulating at zero. The evidence for this is the decay of the *resolved* part of the spectrum, not a computed condition number: at $\sim 10^{18}$ the latter exceeds $1/\epsilon_{\text{mach}} \approx 4.5 \times 10^{15}$, so it reports that $\sigma_{\min}$ is unresolvable and carries no further information — an earlier revision quoted its variation across δ as evidence of flatness, which it cannot be. What the resolved spectrum does show is that σ_i/σ_1 converges pointwise as M, L grow, so $\mathcal{A}$ is a genuine limit object rather than an artifact of

truncation, with the leading ratios still drifting at $M = L = 160$ (for $i = 2$: 0.352, 0.318, 0.272, 0.254 at $M = L = 20, 40, 80, 160$) while the tail fills in from below; and that the converged decay is algebraic rather than geometric, evidenced at large i ($\sigma_{14}/\sigma_2 = 0.14 \approx 2/14$ at $M = L = 160$, against $0.78^{12} = 0.05$ for the geometric rate the first few ratios alone would also fit).

An earlier revision of this section reported a law $\text{rank } \mathcal{A} \approx 0.75 k_M r_c + 2$ and attributed the resolvable pressure order to the bath truncation. That law is withdrawn. Its supporting study set $M = L$ throughout, in which $k_M r_c$ and the droplet-side $L\theta_c$ are collinear, so it had no power to distinguish the two blocks; and it held n_q fixed, so its largest cases were bounded by $\text{rank} \leq n_q$ rather than by the operator. With M and L varied independently and n_q refined to independence, *both* truncations are found to contribute, and which one binds depends on the state. Holding M fixed and starving the droplet: at $L = 160$, $\theta_c = 0.4$ the rank is 25 whether M is 10, 40 or 160 — a sixteenfold change in $k_M r_c$ with no effect — while at $M = 160$, $\theta_c = 0.4$ it falls to 15 for both $L = 40$ and $L = 10$. But holding L fixed and enriching the bath: at $L = 10$, $\theta_c = 0.4$ the rank runs 5, 6, 15, 46 for $M = 10, 40, 160, 640$, and 46 exceeds anything the droplet block can supply there ($L - 1 = 9$ modes plus the rank-one centre-of-mass term), so the bath is contributing the remaining 36 dimensions. M is irrelevant only where the droplet term already saturates. Nor is $L\theta_c$ a substitute for $k_M r_c$: $(M, L, \theta_c) = (160, 40, 0.4)$ and $(80, 80, 0.2)$ share $L\theta_c = 16$ and give ranks 15 and 9. No single-parameter collapse survives the decoupling, and none is asserted here; the withdrawn law failed by omitting the droplet term, not because the bath term is absent. The measured rank is in addition bounded above by n_q , so it is reported only where refining n_q (to 80, 160, 240) leaves it unchanged, and it is δ -independent (measured 14, 15, 15, 15 across $\delta = 10^{-2}, 10^{-3}, 10^{-5}, 10^{-7}$). The practical prescription does not require one: the resolvable dimension is a property of the assembled discrete operator, obtained from its singular values at negligible cost relative to a time step, and it varies through an impact as the patch grows. It is therefore *measured* at each step and used to set the spectral filter of §1.6, rather than predicted from a formula.

The pointwise pressure does not converge; whether its moments do is not established here. That the pointwise pressure is not a converged output is robust across every state and mode count tested: with the filter cutoff at 10^{-10} and $M = L = 800$, $\theta_c = 0.16$, $\min_x p$ runs +98, -2.1×10^3 , -1.0×10^6 , -5.0×10^5 at $N = 10, 20, 40, 80$, and similarly at $M = L = 400$ and at $\theta_c = 0.40$. This is the expected behaviour of an ill-posed first-kind inversion and is not by itself a defect: only the moments c_m, b_l, f enter (5), (12) and (14), so only they need converge. An earlier revision claimed that they do. The evidence does not support that claim, and it is withdrawn. Over the same N sequence at $M = L = 800$, $\theta_c = 0.16$, the relative change in b_l between successive N is $1.5 \times 10^{-2}, 3.6 \times 10^{-2}, 6.4 \times 10^{-5}$ — non-monotone — while at $\theta_c = 0.40$ it is $1.9 \times 10^{-2}, 3.3 \times 10^{-2}, 4.4 \times 10^{-2}$, increasing rather than decreasing. Holding N at 40 and sweeping the filter cutoff over 10^{-6} to 10^{-14} moves b_l by $\approx 1.0 \times 10^{-2}$ per decade and f from 11.28 to 11.92: the cutoff sensitivity is the same size as the N -drift, so the two cannot be separated by this data. And f itself differs by 7% between $M = 400$ and $M = 800$ at fixed $\theta_c = 0.16$ (12.55 against 11.64), so convergence in M is not established either. The honest status is therefore that the closure’s conditioning and structure are established (§1.5.6, §1.6) while its convergence is not; establishing it requires a refinement study in N, M, L and cutoff jointly, at admissible states, which is not carried out here.

1.5.6 The contact angle: an outer selector, not a joint unknown

The single remaining unknown is $\theta_c(\tau)$. Write

$$C(\theta, \tau) := \eta(r(\theta, \tau), \tau) - z_{cm}(\tau) + z_d(\theta, \tau) \quad (34)$$

for the pointwise contact mismatch, which (26) requires to vanish for $0 \leq \theta \leq \theta_c(\tau)$ and, by (38) below, to remain negative beyond it.

Tangency, and the sense in which it is degenerate. The classical closing condition for a free boundary of this kind is tangency — smooth pasting between the pressed and unpressed regions, $\partial_\theta C(\theta_c, \tau) = 0$. It carries an exact degeneracy at the poles. By (1), $\xi(\theta, \tau)$ depends on θ only through $\cos \theta$, hence is even in θ for *every* θ , not merely near the axis; $r(\theta, \tau) = \xi(\theta, \tau) \sin \theta$ is therefore odd; $\eta(r(\theta, \tau), \tau) = \sum_m a_m(\tau) J_0(k_m r)$ is even, since J_0 is an even function of its (here, odd-in- θ) argument, so the composition is even; and $z_d(\theta, \tau) = \xi(\theta, \tau) \cos \theta$ is even $\times$ even, hence even. Every formula composing $C(\theta, \tau)$ ($\cos \theta$, $\sin \theta$, J_0 , the finite Legendre sum in ξ) is entire and defined for every real θ , so this parity statement is a genuine global identity about the analytic extension of C to all of $\mathbb{R}$, not an assumption smuggled in by extending the physical domain. C is therefore even in θ at every θ and every physical state, its θ -derivative is odd and 2π -periodic, and an odd 2π -periodic function vanishes at both fixed points of $\theta \mapsto -\theta$:

$$\partial_\theta C(0, \tau) \equiv 0, \quad \partial_\theta C(\pi, \tau) \equiv 0 \quad (35)$$

identically, for *any* a_m, β_l whatsoever. Near $\theta = 0$, $\partial_\theta C(\theta) = C''(0)\theta + O(\theta^3)$ is affine through the origin for any state, so a Newton iteration on $\partial_\theta C = 0$ seeded anywhere near onset finds the trivial root essentially independent of the physics — confirmed directly: joint solves seeded at $\theta_c^{(0)} \in \{0.02, \dots, 0.5\}$, at various levels of accumulated contact history, all converged to $\theta_c \sim 10^{-8}$.

An earlier revision of this document read (35) as showing tangency to be *structurally* unusable, and discarded it. That inference was too strong, and is withdrawn (§3). Equation (35) is a statement about the poles only, and it is the correct *physical* degeneracy there: at $\theta_c = 0$ the contact patch has shrunk to a point and tangency carries no information about its extent, because there is no extent. The identical degeneracy holds in Agüero et al. (2026)’s own formulation — axisymmetry forces $\partial_r \eta(0, t) = \partial_r z_d(0, t) = 0$ for every state, so their tangency condition (Agüero et al., 2026) is likewise vacuous at $r_c = 0$ — and they use it successfully throughout an impact regardless. The difference is not in the condition but in how it is used: they never solve it by a root-finder seeded at an arbitrary contact extent. They evaluate its residual at a handful of candidate contact extents bracketing the previous step’s, reject those violating non-intersection outright, and take the argument minimising the residual among the survivors. Tangency is serviceable as a *selector* over candidate contact extents; it is degenerate only as an *equation solved from a free seed*. What follows adopts the former role.

The nesting, and why it is not circular. Let θ_c be an outer scalar *parameter* rather than an unknown carried alongside the pressure. For each trial value, $x_c = \cos \theta_c$ is fixed, the pressure representation (25) and every projection of §1.5.3–§1.5.4 are well defined, and the $N + 1$ Galerkin conditions (27) form a square system for the $N + 1$ coefficients $\hat{c}_n$ alone. Solving it yields a pressure, hence — through (44) — a complete state, hence a scalar residual in θ_c with the pressure already eliminated. The outer problem is one-dimensional.

An earlier revision asserted that computing θ_c this way is circular, on the grounds that $p(x; X)$ ’s support depends on θ_c , so b_l, c_m, a_m, β_l cannot be evaluated without θ_c already known. That argument is correct, and it rules out the *opposite* nesting — computing θ_c from an already-determined $\hat{c}$, downstream of quantities that presuppose it. It says nothing against the nesting above, in which θ_c is supplied *first*, as a parameter, and everything downstream is evaluated from it in the same directed acyclic order §1.6 already establishes. Conflating the two directions was the error; §3 records it.

The selectors, and their agreement. Two scalar conditions are available to fix θ_c once the

pressure is eliminated, and they are not independent. The first is tangency in its selector role,

$$T(\theta_c, \tau) := \partial_\theta C(\theta, \tau)|_{\theta=\theta_c}, \quad \text{sought} = 0, \quad (36)$$

evaluated with the state induced by the pressure that (27) returns at that same θ_c . The second is the non-intersection condition (38), imposed not as an afterthought but as a feasibility filter on candidate θ_c exactly as Agüero et al. (2026) impose it: any trial value for which $C(\theta, \tau) \geq 0$ for some $\theta > \theta_c$ is inadmissible and discarded before T is consulted at all.

How well these two locate the same free boundary is a numerical question, and the answer is weaker than an earlier revision of this section claimed. At the representative penetrating state of `derivations/audit_nested_closure.jl` ($M = L = 80$, $\delta = 10^{-3}$, centre of mass at $\mu = 0.985$, so the undeformed surfaces cross at $\arccos \mu = 0.1734$), scanning θ_c on a grid of step 0.005 gives

	$T = 0$	(38) becomes feasible	$\min_x p$ changes sign
$N = 1$	0.1925	0.1725	0.1975
$N = 2$	0.1925	0.1675	0.2025
$N = 3$	0.1925	0.1625	0.1125

The three do *not* coincide — they spread over 0.025 to 0.080 depending on N , and the positivity crossing is erratic at $N = 3$ — so the earlier claim that they agree on one interval, and the inference that the closure was thereby exhibiting a complementarity structure, are both withdrawn (§3). What the table does show is more useful for the closure’s actual purpose: the tangency root is 0.1925 at *every* N , identically to four figures, so the selector is insensitive to the pressure truncation; and the non-intersection boundary sits within 1% of the parameter-free geometric estimate $\arccos \mu$, with the tangency root about 11% above it. Pressure positivity is not imposed anywhere in this closure — consistent with both parent models, neither of which constrains the sign of p pointwise (Alventosa, Cimpeanu & Harris (2023) monitor only sign f ; Agüero et al. (2026)’s constraint list contains no positivity condition) — and it holds on roughly the same interval without being asked to.

The selector is an argmin over candidates, not a bisection, and not an integrated residual. θ_c^{k+1} is fixed exactly as Agüero et al. (2026) fix their contact-point count: evaluate the residual at a finite set of candidate values bracketing the previous step’s converged θ_c^k , discard those failing the feasibility filters (38) and (39), and take

$$\theta_c^{k+1} = \arg \min_{\theta \in \Theta^{k+1}} |T(\theta, \tau^{k+1})|, \quad (37)$$

Θ^{k+1} the admissible candidates. Two properties of this construction are not incidental. It needs no sign change, so the measured non-monotonicity of T — which does occur, and which can place two roots inside one bracket — does not defeat it, whereas a bisection on T would require a bracket guaranteed to contain exactly one root and no such guarantee is available. And T is a *pointwise* quantity evaluated at the contact edge, of fixed size as θ_c varies.

That last point is a genuine constraint on the design, established by trying the alternative. A selector minimising an *integrated* weak residual over the contact patch — for instance the least-squares residual of a system combining the kinematic conditions on $[x_c, 1]$ with weakly-imposed zero pressure on $[-1, x_c]$, using a pressure basis global on $[-1, 1]$ — is biased toward vanishing contact, because shrinking the patch shrinks the domain the residual is measured over. Measured (`derivations/probe_global_basis_argmin.jl`): that residual is monotone increasing away from $\theta_c = 0$, so its argmin is the smallest admissible θ_c , reproducing the onset degeneracy rather than curing it. The same probe records two further obstructions to the global-basis route: the number

of test functions imposing zero pressure off the patch must be strictly less than $N + 1$, since $N + 1$ of them form a square full-rank system admitting only $p \equiv 0$ (the identity-theorem obstruction in weak form); and with fewer, the combined system is numerically rank deficient, $\text{cond} \sim 10^{14}$ to 10^{22} , so its least-squares residual sits at roundoff for every θ_c and carries no signal at all. Agüero et al. (2026)’s argmin is not subject to any of this precisely because their residual is a pointwise edge quantity; (37) follows them.

Conditioning. Removing θ_c from the linear system is what repairs the conditioning, and the mechanism is now unambiguous. By §1.5.5, every column of the pressure block carries a factor of κ_m , κ_{cm} or λ_l , all $O(\delta^2)$; a θ_c column does *not*, since θ_c is a raw coordinate that never flows through the BDF2 relations (22)–(24). A joint system therefore mixes columns differing by δ^2 in scale, and its condition number grows as δ shrinks — exactly the measured behaviour of the joint formulation. The inner system of the nesting contains no such column, and its condition number is measured (`derivations/audit_nested_closure.jl`, $N = 3$, $M = L = 80$, $\theta_c = 0.3$) at 8.19, 3.11, 2.97, 2.96, 2.96 for $\delta = 10^{-2}, \dots, 10^{-6}$: flat in δ and of order unity, against $\sim 10^{18}$ for the joint system. The gain is roughly eighteen orders of magnitude and is obtained by removing one row, not by adding machinery.

Bracketing, in place of a multiplicity argument. Because $\eta(r(\theta, \tau), \tau)$ is a sum of oscillatory J_0 terms once $k_m r$ is not small, $T(\cdot, \tau)$ may change sign more than once on $(0, \pi)$, and $C(\cdot, \tau)$ likewise. This is the same multi-rootedness a joint Newton solve would face, but it is far more tractable in one dimension: the outer problem admits bracketing. Following Agüero et al. (2026), the search is confined to a neighbourhood of the previous step’s converged θ_c^k and the accepted value is required to satisfy a continuity bound $|\theta_c^{k+1} - \theta_c^k|$ below a threshold, with the time step reduced and the step retried when it is not — a physical requirement (the contact line cannot jump) rather than a numerical expedient, and the direct analogue of their $|r_c^{k+1} - r_c^k| \leq \delta_r$. Within such a bracket (37) selects among finitely many candidates, which cannot return a value outside the bracket.

Onset — not specified here. One step is not covered by the construction above, and the gap is stated rather than papered over. At first contact $\theta_c^k = 0$, and (35) makes $T \equiv 0$ identically there for every state, so T is uninformative in a neighbourhood of $\theta_c = 0$ and the first bracket cannot be centred on the previous step’s value. The scanned data show the degeneracy reaching into the working range rather than staying at the endpoint: T is non-monotone and decreasing toward the bottom of the admissible window. Agüero et al. (2026) guard this explicitly — their discrete tangency condition is stated “if $q > 0$ ”, and their candidate loop runs over integers so that $q = 0$ is a distinguishable, excluded state — and this document has no analogue of either guard. A pragmatic initialisation is available and is what the implementation uses: at the first contact step, seed θ_c from the geometric crossing $C(\theta_c, \tau) = 0$, which is non-degenerate at onset (§1.5.6’s existence argument) and which the table above places within 1% of the reference, then hand over to (37) from the second step on. That is an initialisation choice, not a derived rule, and it is recorded as such.

Existence of at least one admissible θ_c for a state genuinely in contact follows as before from the intermediate value theorem applied to C : $C(0, \tau) > 0$ is the geometric onset-of-contact condition this closure presupposes (the pole’s absolute height is $z_{cm} - \xi(0, \tau)$, so the gap there is $-C(0, \tau)$, negative exactly when the pole has penetrated), while $C(\theta, \tau)$ must become negative before $\theta = \pi$, since a positive C all the way to π would mean the surfaces overlap everywhere — not a state this document’s kinematic-match premise admits. Continuity of C in θ then gives a crossing. This is a plausibility check on the closure, not a guarantee that the bracketed search finds the physically continuous branch at every step; that remains a numerical matter, monitored as described.

1.5.7 Admissibility

Two conditions are structural in the closure above, and one is a diagnostic rather than a constraint. Non-intersection of the two surfaces beyond the contact edge,

$$\eta(r(\theta, \tau), \tau) < z_{cm}(\tau) - z_d(\theta, \tau), \quad \theta \in (\theta_c(\tau), \pi], \quad (38)$$

is a feasibility filter on candidate θ_c , applied before the selector (36) is consulted, exactly as in Agüero et al. (2026). The kinematic match itself holds in the weighted-residual sense (27) on the contact patch, and $p \equiv 0$ off it holds exactly by construction of (25).

A second condition is structural and was left implicit in an earlier revision. Every substitution of $r dr = w dx$ in (29), (32) and (47) requires $x \mapsto r(x, \tau)$ to be injective on the patch, equivalently

$$\frac{\partial r}{\partial x}(x, \tau) < 0, \quad x \in [x_c(\tau), 1], \quad (39)$$

without which $w = -\frac{1}{2}d[r^2]/dx$ is not $r|dr/dx|$, can change sign, and the change of variables is invalid. For an undeformed droplet $r = \sqrt{1 - x^2}$ and (39) holds throughout; for a deformed one it fails once the patch reaches the widest point of the droplet, which is exactly Agüero et al. (2026)’s $r_M(t)$ — “the smallest distance from the z axis to the point on the surface of the droplet for which the plane that is tangent to the interface of the droplet is vertical”, which they introduce precisely as an upper bound on the admissible contact radius. Condition (39) is the same bound, $r_c(\tau) < r_M(\tau)$, and must be tested on each candidate θ_c alongside (38).

Positivity of the pressure throughout the pressed region,

$$p(x, \tau) = \sum_{n=0}^N \hat{c}_n(\tau) \tilde{P}_n(\psi(x, \tau)) \geq 0, \quad x \in [x_c(\tau), 1], \quad (40)$$

is neither imposed nor, at large N , a meaningful test. It is not imposed because neither parent model imposes it and because it is observed to hold on the admissible interval without being asked to (§1.5.6). It ceases to be a meaningful test at large N for the reason established in §1.5.5: the compliance operator is compact, so the pointwise pressure is not a converged output of the model at any M , and $\min_x p$ is measured to diverge with N . A pointwise sign test applied to a quantity the model determines only weakly reports the truncation, not the physics. At small N , where the pressure is close to fully resolved, (40) remains informative and is worth monitoring; at larger N the corresponding diagnostic is sensitivity of c_m, b_l, f to the spectral-filter cutoff, since it is the cutoff and not N that sets the resolution there (§1.6). Neither diagnostic is a convergence proof: §1.5.5 records that convergence of the moments is not established.

Nor does (25) constrain the value of the pressure at the edge of contact: $p(x_c, \tau) = \hat{c}_0(\tau) - \hat{c}_1(\tau) + \dots$ is generally nonzero, admitting a jump in p across $\theta_c(\tau)$. This is not obviously a defect. Alventosa, Cimpeanu & Harris (2023)’s review of direct numerical simulation and experiment on this problem (not solid–solid Hertzian contact) finds the true pressure distribution during impact closer to flat than singular at the edge, which is why they fit a high-order rather than singular shape function — so a prescribed vanishing at θ_c is not evidently the right target, and the edge behaviour is a property of the solution rather than a constraint to impose in advance.

1.6 Solution procedure

The contact problem at time level τ^{k+1} is solved as a one-dimensional outer iteration over θ_c^{k+1} wrapping an inner solve for the pressure coefficients. The inner unknowns are

$$\hat{c} := (\hat{c}_0^{k+1}, \dots, \hat{c}_N^{k+1}) \in \mathbb{R}^{N+1}, \quad (41)$$

with θ_c^{k+1} — and hence $x_c^{k+1} = \cos \theta_c^{k+1}$ — held fixed at the current trial value throughout. Every other quantity is an explicit function of $\hat{c}$, computed in the following order with no step depending on a quantity not yet available: $p(x; \hat{c})$ from (25), using the fixed x_c^{k+1} ; $b_l(\hat{c})$ and the droplet contribution to (27) from (13) and (28); $\beta_l(\hat{c}) = \gamma_l^{k+1} + \lambda_l b_l(\hat{c})$ from (23); $\xi(x; \hat{c}) = 1 + \sum_l \beta_l(\hat{c}) P_l(x)$, $r(x; \hat{c}) = \xi \sqrt{1 - x^2}$, $z_d(x; \hat{c}) = \xi x$ from (1)–(2); $c_m(\hat{c})$ and $W_n^{(m)}(\hat{c})$ from (29); $a_m(\hat{c}) = \alpha_m^{k+1} + \kappa_m c_m(\hat{c})$ from (22); and $f(\hat{c})$, $z_{cm}(\hat{c}) = \mu^{k+1} + \kappa_{cm} f(\hat{c})$ from (14), (24). The inner residual is

$$R_n(\hat{c}) := \int_{x_c}^1 \left[\eta(r(\theta; \hat{c}), \tau^{k+1}) - z_{cm}(\hat{c}) + z_d(\theta; \hat{c}) \right] \tilde{P}_n(\psi(x)) dx, \quad n = 0, \dots, N, \quad (42)$$

$N + 1$ equations for $N + 1$ unknowns, solved by Newton’s method

$$\hat{c}^{(j+1)} = \hat{c}^{(j)} - J(\hat{c}^{(j)})^{-1} R(\hat{c}^{(j)}), \quad J := \frac{\partial R}{\partial \hat{c}}. \quad (43)$$

The system is genuinely nonlinear, though only mildly: by §1.5.5 it would be exactly linear, $R = \langle \tilde{P}_n, C_{\text{free}} + \mathcal{A}\hat{c} \rangle$, were the geometry frozen; the nonlinearity enters entirely through the $\beta_l(\hat{c})$ -dependence of $r(x; \hat{c})$: non-polynomially inside $J_0(k_m r)$, and *quadratically* through $r^2 = \xi^2(1 - x^2)$, which appears in f and in w and therefore makes $z_{cm}(\hat{c})$ a quadratic rather than affine function of $\hat{c}$. Both channels are $O(\delta^2)$ -suppressed by (23), which is why Newton converges rapidly from the previous time level’s converged $\hat{c}$ — an expectation, not yet a measurement. What has been measured is the conditioning of the frozen-geometry Galerkin matrix $\langle \tilde{P}_n, \mathcal{A}\tilde{P}_{n'} \rangle$ that J reduces to when the geometry is held fixed: of order unity and flat in δ (§1.5.6), provided $N + 1$ does not exceed the measured resolvable dimension of §1.5.5. Neither $\text{cond}(J)$ at a converged nonlinear state nor the Newton iteration count has been measured here. Because every arrow in the chain above (polynomial evaluation, Gauss–Legendre quadrature (30), and J_0, J_1 with their derivative recurrences $J'_0 = -J_1$, $J'_1 = J_0 - J_1/x$) is an elementary or special-function composition, and none is a linear solve, an implicit root or a search, R is differentiable by direct automatic differentiation of the same computation that evaluates it; forming J costs $N + 1$ forward-mode passes and requires no adjoint rule.

Equation (43)’s linear solve is always performed as a spectral-filtered solve — a truncated singular value decomposition of J at a fixed relative cutoff, discarding directions below it — and not only when N is large. This is the standard treatment of a compact operator’s ill-posed inversion rather than a tuning knob in disguise: by §1.5.5 the discarded directions are those in which the pressure does not measurably affect the gap, and the retained count is exactly the measured resolvable dimension, which adapts on its own as the patch grows through the impact. Adopting it unconditionally is what makes the policy consistent: the resolution is set by the cutoff, so N may be over-provisioned freely, and no rule for choosing N correctly is needed. The price is that the cutoff is now the parameter the solution depends on, and §1.5.5 records that this dependence is of the same size as the residual N -dependence, so it must be reported rather than assumed negligible. Choosing N strictly inside the resolvable dimension, so that no filtering is active, remains available and is the regime in which $\hat{c}(\theta_c)$ is unambiguously single-valued.

The outer iteration is then a one-dimensional search for θ_c^{k+1} : bracket around θ_c^k subject to the continuity bound, discard trial values violating (38), and bisection on the sign change of (36) among the survivors, reducing δ and retrying the step if no admissible bracket exists. Each outer evaluation costs one inner solve. The scheme is fully implicit in the pressure and requires no extrapolated state anywhere.

1.7 Model summary

The complete system solved at every time level τ^{k+1} : the $N + 1$ equations (48) for $\hat{c}^{k+1}$ at fixed θ_c^{k+1} , wrapped in the scalar selection (49) for θ_c^{k+1} itself, together with the states, geometry and pressure moments that §1.6 shows to be explicit functions of $\hat{c}^{k+1}$ alone.

States, advanced from (22)–(24) given the pressure moments at τ^{k+1} :

$$a_m^{k+1} = \alpha_m^{k+1} + \kappa_m c_m^{k+1} \quad (m = 0, \dots, M), \quad \beta_l^{k+1} = \gamma_l^{k+1} + \lambda_l b_l^{k+1} \quad (l = 2, \dots, L), \quad z_{cm}^{k+1} = \mu^{k+1} + \kappa_{cm} f^{k+1}. \quad (44)$$

Geometry, given the state:

$$r(\theta, \tau^{k+1}) = \xi(\theta, \tau^{k+1}) \sin \theta, \quad z_d(\theta, \tau^{k+1}) = \xi(\theta, \tau^{k+1}) \cos \theta, \quad \xi = 1 + \sum_l \beta_l^{k+1} P_l(\cos \theta). \quad (45)$$

Pressure, the inner unknowns, at the current trial contact angle:

$$p(x, \tau^{k+1}) = \sum_{n=0}^N \hat{c}_n^{k+1} \tilde{P}_n(\psi(x, \tau^{k+1})) \quad (x \geq x_c^{k+1}), \quad p \equiv 0 \quad (x < x_c^{k+1}), \quad x_c^{k+1} = \cos \theta_c^{k+1}. \quad (46)$$

Pressure moments feeding (44), computed from (46) via (13), (28) and (29), all by quadrature (30):

$$c_m^{k+1} = \frac{2}{(b J_0(k_m b))^2} \int_{x_c^{k+1}}^1 p J_0(k_m r) r \left| \frac{dr}{dx} \right| dx, \quad b_l^{k+1} = \int_{x_c^{k+1}}^1 p x P_l(x) w dx, \quad f^{k+1} = - \int_{x_c^{k+1}}^1 p d[r^2]. \quad (47)$$

Inner closure, $N + 1$ equations for the $N + 1$ unknowns $\hat{c}_0^{k+1}, \dots, \hat{c}_N^{k+1}$ at fixed θ_c^{k+1} :

$$\int_{x_c^{k+1}}^1 \left[\eta(r(\theta, \tau^{k+1}), \tau^{k+1}) - z_{cm}^{k+1} + z_d(\theta, \tau^{k+1}) \right] \tilde{P}_n(\psi(x, \tau^{k+1})) w dx = 0, \quad n = 0, \dots, N, \quad (48)$$

Outer closure, selecting θ_c^{k+1} among candidates admissible under (38), (39) and the continuity bound:

$$\theta_c^{k+1} = \arg \min_{\theta \in \Theta^{k+1}} |\partial_\theta C(\theta, \tau^{k+1})|, \quad C(\theta, \tau) = \eta(r(\theta, \tau), \tau) - z_{cm}(\tau) + z_d(\theta, \tau). \quad (49)$$

The pressure and the contact extent are both solved for — neither is prescribed — but they are solved at different levels rather than jointly, which is what keeps the inner system's conditioning independent of δ and of order unity. η , ξ and p remain finite spectral sums throughout; (2) is used only forward; $p \equiv 0$ off the contact patch holds exactly; and the resolvable pressure order is measured from the assembled compliance operator at each step (§1.5.5) rather than chosen or predicted.

2 Validation

The closure of §1 has been implemented and run against published data for the water case of Alventosa, Cimpeanu & Harris (2023) ($We = 1.0958$, $Bo = 0.017$, $Oh = 0.006$), using the experimental and direct-numerical-simulation records that accompany Agüero et al. (2026) rather than Alventosa, Cimpeanu & Harris (2023)'s own reduced curves, since the former supersede the latter. At $M = L = 60$, $N = 3$, $n_q = 40$, $\delta_{\text{init}} = 10^{-3}$:

series	n	mean $ \text{err} $	max $ \text{err} $	RMS	RMS/bar
droplet top	60	0.1345	0.2503	0.1526	1.52
droplet bottom	39	0.1055	0.2580	0.1221	1.22
combined	99	0.1231	0.2580	0.1414	1.41

against a measurement uncertainty whose median half-bar is 0.1004 (quartiles 0.0810 and 0.1179, from 293 digitised bar caps). The model tracks the measured trajectory at about 1.4 times the experimental error bar — the same order as the measurement scatter, but outside it, so the residual is model error rather than noise. The trajectory is physically bounded and rebounds: the centre of mass enters contact at $v = -1.047$ and leaves at $v = +0.314$.

What can *not* be validated this way must be said plainly, because an earlier revision of this section reported it as though it could. The dataset contains no measured contact radius and no measured droplet width; both exist only as DNS and 1PKM series, verified by reading the digitiser projects that generated them, each of which holds exactly two datasets. Any statement about contact radius is therefore a comparison against another computation rather than against measurement, and is additionally sensitive to definition: the DNS contact radius is thresholded on the trapped gas film, whereas θ_c here bounds the kinematically matched region, and the two need not coincide.

With that caveat the contact-radius behaviour is still worth recording. Against DNS the model’s r_c peaks at $\tau \approx 0.89$ against 1.46, and refining M and L moves that by only 0.02, so it is not a resolution effect. But the early peak is partly a property of this class of reduced model rather than of this model alone: 1PKM peaks at $\tau \approx 1.22$, also early against the same DNS. Normalised by contact time, the peak falls at 0.20 of the way through contact here, 0.26 for 1PKM and 0.33 for DNS — so this model is more skewed than 1PKM, in the same direction, with no measurement available to adjudicate. Whether the remainder is a defect of the closure or a mismatch of definitions is open.

Two discrepancies are larger than the rest and are recorded rather than explained. The contact radius is 7% low at its maximum, and — more tellingly — reaches that maximum at $\tau = 0.87$ against the DNS value $\tau = 1.46$, so the modelled contact patch both grows and recedes too quickly even though the total contact time is accurate to 3%. Whether this is a resolution effect or a defect of the closure is not settled here; it is the natural first target for the next iteration.

A resolution study sharpens §1.5.5’s weak-determinacy finding, and does so on a real trajectory rather than the synthetic state used there. Sweeping the pressure truncation at $M = L = 60$, the contact time is identical to six significant figures at every one of $N = 2, 3, 6, 10, 16$ — while the solved pressure profile at $N = 3$ differs from that at $N = 16$ by 46 to 96% in relative L^2 over the patch, and the net force by between 10^{-5} and 10^{-2} . Both statements are correct and they are not in tension: the pressure acts on the dynamics only through c_m , b_l and f , and the modes that distinguish $N = 3$ from $N = 16$ lie in the numerical nullspace of $\mathcal{A}$. The practical consequence is that trajectories may be computed at $N = 3$, but the pointwise pressure must not be quoted as a converged output at any N — which is the same conclusion §1.5.5 reaches from the operator’s spectrum, now confirmed dynamically.

One observable does drift with N : the maximum contact radius falls monotonically from 0.8192 at $N = 2$ to 0.7999 at $N = 16$, moving *away* from the DNS value 0.8821. That the single quantity most sensitive to the pressure profile is also the one carrying the open timing discrepancy is unlikely to be a coincidence, and is the most concrete lead available for it. Convergence is judged here on contact time rather than the coefficient of restitution deliberately: the latter settles far earlier than anything else and would have reported convergence at mode counts where the contact time is still wrong by a factor of two ($M = L = 30$ gives 2.15 against the converged 4.26).

One implementation defect worth recording, because its symptom was easy to mistake for a cost scaling rather than a bug. Loss of contact was originally declared as soon as the net force f turned non-positive, following Alventosa, Cimpeanu & Harris (2023)’s own sign f criterion. That criterion is too eager here: f can dip transiently negative during the early transient without contact ending — measured at $We = 0.3$, where it runs $+0.0002, +0.0063, +0.0367, -0.0038, +0.0321, +0.0269, \dots$ — and acting on the single excursion returns the stepper to free flight, where onset is immediately re-detected and a fresh onset search is paid. Repeated, the cycle cost that Weber number some twenty times the wall clock of its neighbours (555 s against 23–32 s), which looked like a scaling pathology and was not. Requiring $f \leq 0$ on several consecutive steps removes it: $We = 0.3$ now runs in 24 s, and the contact time becomes monotone decreasing in We across the range tested, $We = 0.0646$ to 3.0, as it should be. The complementary failure is also real and is guarded separately: as the patch collapses $f \rightarrow 0$ from above and never changes sign, so the primary loss-of-contact test is the contact angle reaching its search floor, not the force.

The implementation exercises exactly the structure derived above, and two of its details were forced by measurement rather than chosen. First, reducing the time step does *not* help a step whose candidates all fail the non-intersection filter: the pressure’s ability to deform either interface within a step scales as δ^2 (§1.5.5), so a smaller step has strictly less authority to remove an overlap, and the usual shrink-and-retry response is counterproductive here. Second, the feasible set of contact angles is a *band*, not a half-line — infeasible below, where the surfaces interpenetrate beyond the edge, and infeasible again above, where (39) fails once the patch reaches the droplet’s widest point — so the lower edge must be bracketed by a search that starts from the previous step’s angle and walks outward. A plain bisection assuming monotonicity walks past the band and reports no admissible angle at steps where one exists.

3 Corrections to earlier revisions

This document has been revised substantially, and several arguments in earlier revisions were wrong rather than merely superseded. They are recorded here because each was used to justify structure that has now been removed, and because two of them were load-bearing for an entire section.

The $O(\delta^2)$ conditioning diagnosis, and the acceleration-level closure built on it. An earlier revision contained a section arguing that the position-level closure (27) *could not* be well conditioned, because η, z_{cm}, z_d respond to the current step’s pressure only at $O(\delta^2)$; it proposed replacing (27) by a Galerkin projection of $\ddot{C} = 0$, obtained by differentiating the contact condition twice in τ and substituting the governing ODEs, so that the pressure would enter at $O(1)$. The premise is false, though not for the reason a previous revision of *this* section gave. That revision asserted the $O(\delta^2)$ factor to be exactly *uniform* across the pressure block; §1.5.5 now measures the factorisation $\mathcal{A} = \delta^2 \hat{\mathcal{A}}$ to be asymptotic only, failing badly (96% deviation in κ_m/δ^2) at the mode counts of interest. The refutation does not need uniformity. It needs only that a scalar factor cannot change a condition number, $\text{cond}(c\mathcal{A}) = \text{cond}(\mathcal{A})$, together with the direct measurement that the inner system’s condition number is flat in δ and of order unity once θ_c is removed from it. The measured δ -dependence of the joint system’s Jacobian came from mixing the $O(\delta^2)$ pressure columns with a θ_c column carrying no such factor; removing θ_c from the linear system, as §1.5.6 now does, leaves a condition number flat in δ and of order unity. The acceleration-level section has accordingly been deleted in full, along with the open “hidden constraint drift” problem it introduced — an artifact of differentiating a constraint that did not need differentiating, and not a defect of the underlying model. The residual ill-conditioning that section attributed partly to the pressure basis

is now identified as the compactness of $\mathcal{A}$ (§1.5.5): δ -independent, basis-independent in origin, and correctly handled by spectral filtering rather than by raising the differential order of the closure.

Tangency as “structurally unusable.” The degeneracy (35) is real, but the inference that tangency cannot serve as a closing condition was too strong. It is a statement about the poles, where it is the correct physical degeneracy, and it holds identically in Agüero et al. (2026), who use tangency throughout an impact nonetheless. What (35) rules out is solving tangency by a root-finder seeded at an arbitrary contact extent, not tangency itself. §1.5.6 uses it in the selector role its degeneracy permits.

The circularity argument. The claim that determining θ_c from the pressure is circular is correct for the nesting in which θ_c is computed *after* $\hat{c}$, and was mistakenly applied to the nesting in which θ_c is supplied first as a parameter. The two are not the same, and the latter is not circular.

Piecewise pressure support as a “structural requirement.” The identity-theorem argument of §1.5.1 was used to reject a global pressure representation with weakly-imposed vanishing outside contact. The observation is true but not disqualifying, since it condemns (27) on the same grounds; the piecewise representation is retained on the different and weaker ground that the outer nesting makes it free.

M, L and N as free parameters, and a withdrawn rank law. An earlier revision treated the pressure truncation N as an independent choice and recorded as an unresolved concern that $\text{cond}(J)$ grew sharply with it. A subsequent revision claimed to resolve this with a law $\text{rank } \mathcal{A} \approx 0.75 k_M r_c + 2$, attributing the resolvable pressure order to the bath truncation. That law is withdrawn (§1.5.5): its supporting study set $M = L$, making the claimed collapse variable collinear with a droplet-side one, and held n_q fixed, capping its largest cases at the quadrature count; decoupling M from L shows both truncations to contribute — a ninefold rank swing at fixed L — with no single-parameter collapse surviving. The resolvable dimension is measured from the assembled operator instead. Relatedly, that revision’s claim that the pressure’s moments converge in N while its pointwise values do not is also withdrawn: only the second half is supported, and the residual drift in the moments is the same size as the spectral filter’s own cutoff sensitivity.

The self-adjointness diagnosis, twice wrong in opposite directions. One revision of §1.5.5 stated that the droplet block’s asymmetry is caused by “a single factor of x ”, that repairing $b_l \rightarrow \int p x P_l dx$ removes it, and that the operator is monotone because its symmetric part dominates in Frobenius norm. All three were wrong: the repair was measured in a different inner product from the one the claim was stated in; the two candidate mechanisms are indistinguishable at $\beta = 0$, where $w \equiv x$ exactly, so no measurement taken at the undeformed state could have separated them; and norm dominance of the symmetric part is not evidence of definiteness in either direction. The following revision over-corrected, concluding from the failure of each repair *in isolation* that no mechanism was identifiable and that self-adjointness was unattainable. That is also withdrawn: repairing both mismatches together, (33), makes the whole operator self-adjoint to 10^{-16} , and the formulation now adopts it. What survives unchanged from the first correction is the definiteness statement: the operator is measured positive semidefinite with a large nullspace, giving uniqueness of the gap response and not of the pressure.

The three-selector agreement. A revision reported that the tangency root, the non-intersection feasibility boundary and the pressure-positivity crossing all fall between $\theta_c = 0.16$ and 0.18 at every N , and read this as evidence of a complementarity structure. Those numbers predated the Bessel-normalization correction and do not survive it: on a refined grid the three spread over 0.025 to 0.080 (§1.5.6). The claim is withdrawn. What replaces it is narrower and more directly useful — the tangency root is N -independent to four figures — and the structural inference drawn from the coincidence is retracted with it.

The contact time, reported as a span. Every contact time in earlier revisions of this document,

and in the repository’s README, was computed as the interval from the first contact step to the last. That is wrong, and by more than a rounding: a single impact does not produce a single contact interval. The stepper detaches and re-attaches several times, because the guard that forces one advancing free-flight step after contact ends is followed by immediate re-detection of onset at the next step, so the “span” silently includes both that chatter and any genuine free-flight excursion between re-contacts. At the reference case the three candidate definitions give 3.629 (first interval), 3.815 (sum over intervals) and 4.262 (span), a spread of 17%. The reported metric is now the first interval’s duration, which is the only one measuring a single physical event; the sum remains appropriate for a dissipation budget. Two consequences are worth recording. The span was not merely imprecise but non-monotone in We , whereas the corrected metric decreases monotonically as it should, so the error was masking a diagnostic. And the N -independence claim survives the correction unchanged — the contact time is identical to six significant figures across $N = 2, \dots, 16$ under either definition — so nothing built on that claim is affected.

The case against a constrained no-flux basis for a pinned bath. §1.2.1 argued in an earlier revision that the no-flux set cannot usefully represent a pinned surface, on the strength of a measured wall slope falling six-fold short of the true value at $M = 640$. The diagnostic was measuring its own grid: a finite difference across the last interval of a fixed grid, taken inside a boundary layer of width $\sim b/k_M$, returns a number proportional to $\Delta r k_M$, and refining the grid at fixed M drives it to zero. Computed analytically the truncation error falls as $M^{-3/2}$ and the wall slope is recovered at every $r < b$. The conclusion is reversed accordingly: the constrained route is the recommended one. The multiplier algebra given alongside it was also wrong — the constraint force was inserted with the wrong modal normalization, giving a denominator $D = \sum_m J_0(k_m b)^2 \kappa_m / F_m$ whose $m = 0$ term is $0/0$ and whose quoted values were neither the sum the formula defines nor the correct one. Treating the multiplier as the line force it physically is gives (11), in which the $J_0(k_m b)$ factors cancel. Separately, the implemented pinned option was recommended on the ground that its only failure was a quantified violation of no-flux; that understated it, since the true cost is loss of exact volume conservation, at four times the droplet’s volume.

Bisection, and the integrated-residual selector. A revision specified the outer selector as a bisection on the sign change of T ; another proposed replacing it with an argmin of an integrated weak residual over a global pressure basis. Both are withdrawn in favour of (37). Bisection requires a bracket containing exactly one root, which the measured non-monotonicity of T does not provide, and is undefined at onset. The integrated-residual selector is biased toward vanishing contact, because shrinking the patch shrinks the domain the residual is measured over; measured, its argmin is the smallest admissible θ_c , so it reproduces the onset degeneracy instead of curing it, and the global basis it requires is separately obstructed (§1.5.6). The lesson, stated because it was not obvious in advance, is that Agüero et al. (2026)’s argmin works because its residual is a *pointwise edge* quantity of fixed size, not because it is an argmin.

References

- Alventosa, L. F. L., Cimpeanu, R. & Harris, D. M. 2023 Inertio-capillary rebound of a droplet impacting a fluid bath. *J. Fluid Mech.*
- Agüero, E., Galeano-Rios, C. A., Ragazzo, C., Gabbard, C. T., Harris, D. M. & Milewski, P. A. 2026 Droplet rebounds off a fluid bath at low Weber numbers. *J. Fluid Mech.*
- Dias, F., Dyachenko, A. I. & Zakharov, V. E. 2008 Theory of weakly damped free-surface flows: a new formulation based on potential flow solutions. *Phys. Lett. A* **372**, 1297–1302.

- Graham-Eagle, J. 1983 A new method for calculating eigenvalues with applications to gravity-capillary waves with edge constraints. *Math. Proc. Camb. Phil. Soc.*
- Hocking, L. M. 1987 The damping of capillary-gravity waves at a rigid boundary. *J. Fluid Mech.*
- Scott, J. C. 1979 Gravity-capillary waves with edge constraints. *J. Fluid Mech.*
- Lamb, H. 1932 *Hydrodynamics*, 6th edn. Cambridge University Press.